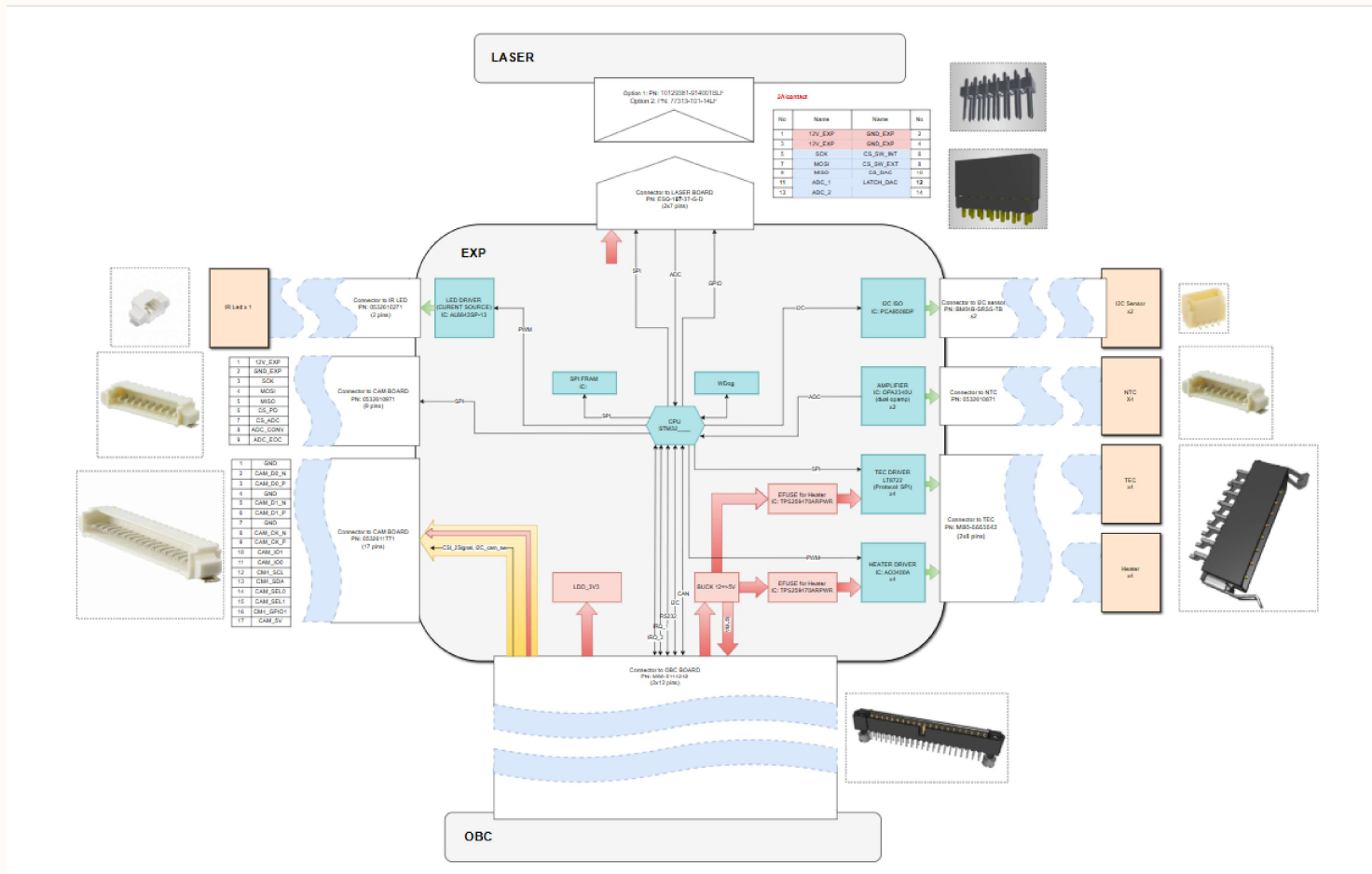
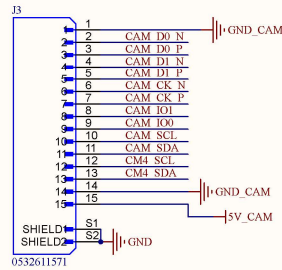
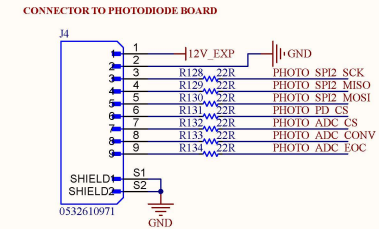
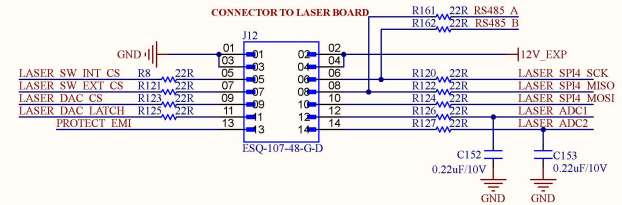
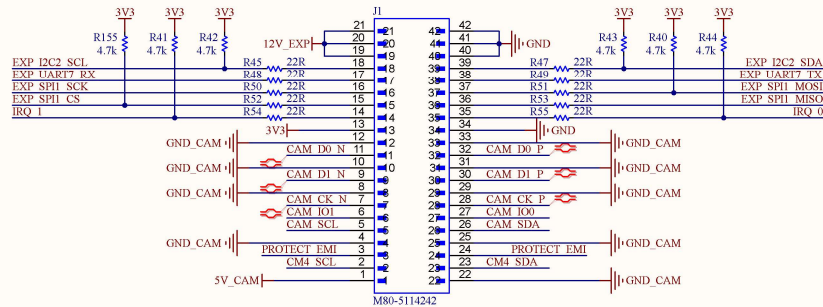


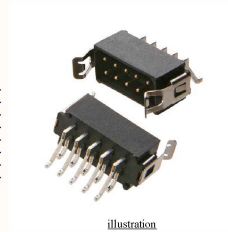
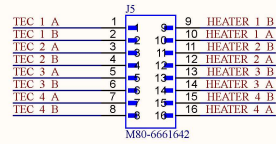


|                               |                                            |                             |
|-------------------------------|--------------------------------------------|-----------------------------|
| Project: <b>BEE-PC1</b>       | Sub-system: <b>NANORACK_EXP_CONTROLLER</b> |                             |
| Design by: <b>SANG HUYNH</b>  | Sheet title: <b>Block_Diagram.SchDoc</b>   | Sheet <b>1</b> of <b>12</b> |
| Checked by: <b>BAO BUI Q.</b> | Document No: <b>1</b>                      | Size: <b>A3</b>             |
| Approved by: <b>VU PHAM</b>   | Revision: <b>V1.2.1</b>                    | Date: <b>4/11/2025</b>      |



|                               |                                            |                             |
|-------------------------------|--------------------------------------------|-----------------------------|
| Project: <b>BEE-PC1</b>       | Sub-system: <b>NANORACK_EXP_CONTROLLER</b> |                             |
| Design by: <b>SANG HUYNH</b>  | Sheet title: <b>HEADER_1.SchDoc</b>        | Sheet <b>2</b> of <b>12</b> |
| Checked by: <b>BAO BUI Q.</b> | Document No: <b>2</b>                      | Size: <b>A3</b>             |
| Approved by: <b>VU PHAM</b>   | Revision: <b>V1.2.1</b>                    | Date: <b>4/11/2025</b>      |

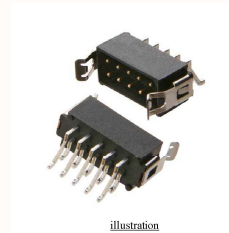
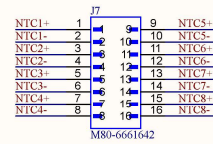
# CONNECTOR TO TEC & HEATER



## Harwin Inc. M80-6661642

- **ELECTRICAL:**  
CURRENT RATING: 2.2A/CONTACT  
VOLTAGE RATING: 800V/CONTACT
- **ENVIRONMENTAL:**  
TEMPERATURE RANGE: -55°C TO +125°C
- **PHYSICAL:**  
PITCH-MATING: 0.079" (2.00mm)
- **WIRE:**  
WIRE SIZE: 24 AWG  
MAX CURRENT: 2A per Wire  
ISULATION MATERIAL: PTFE
- **MATING:**  
Harwin Inc. M80-8881606

# CONNECTOR TO NTC



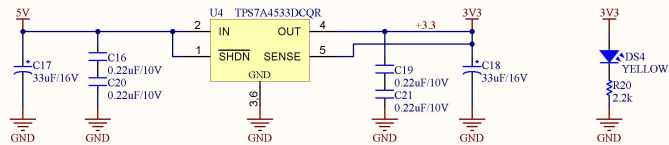
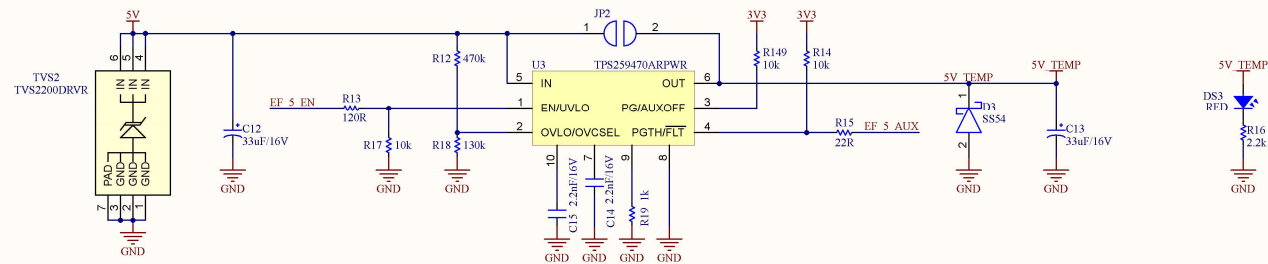
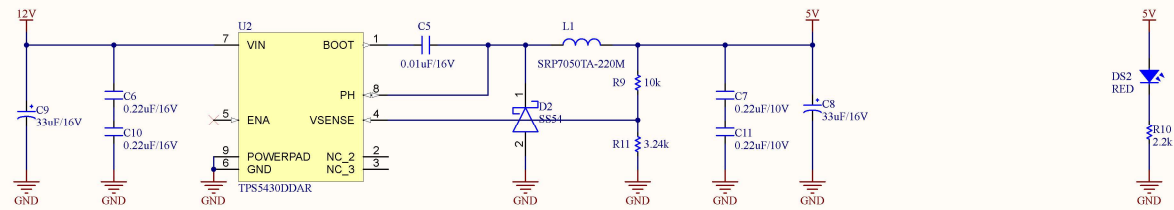
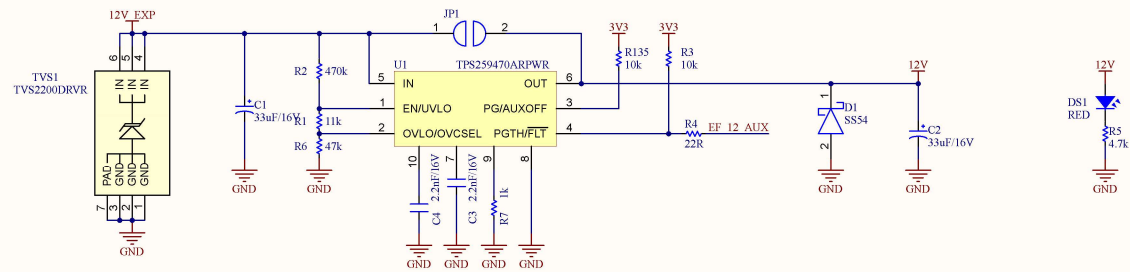
## Harwin Inc. M80-6661642

- **ELECTRICAL:**  
CURRENT RATING: 2.2A/CONTACT  
VOLTAGE RATING: 800V/CONTACT
- **ENVIRONMENTAL:**  
TEMPERATURE RANGE: -55°C TO +125°C
- **PHYSICAL:**  
PITCH-MATING: 0.079" (2.00mm)
- **WIRE:**  
WIRE SIZE: 24 AWG  
MAX CURRENT: 2A per Wire  
ISULATION MATERIAL: PTFE
- **MATING:**  
Harwin Inc. M80-8881606

# MOUNTING HOLE

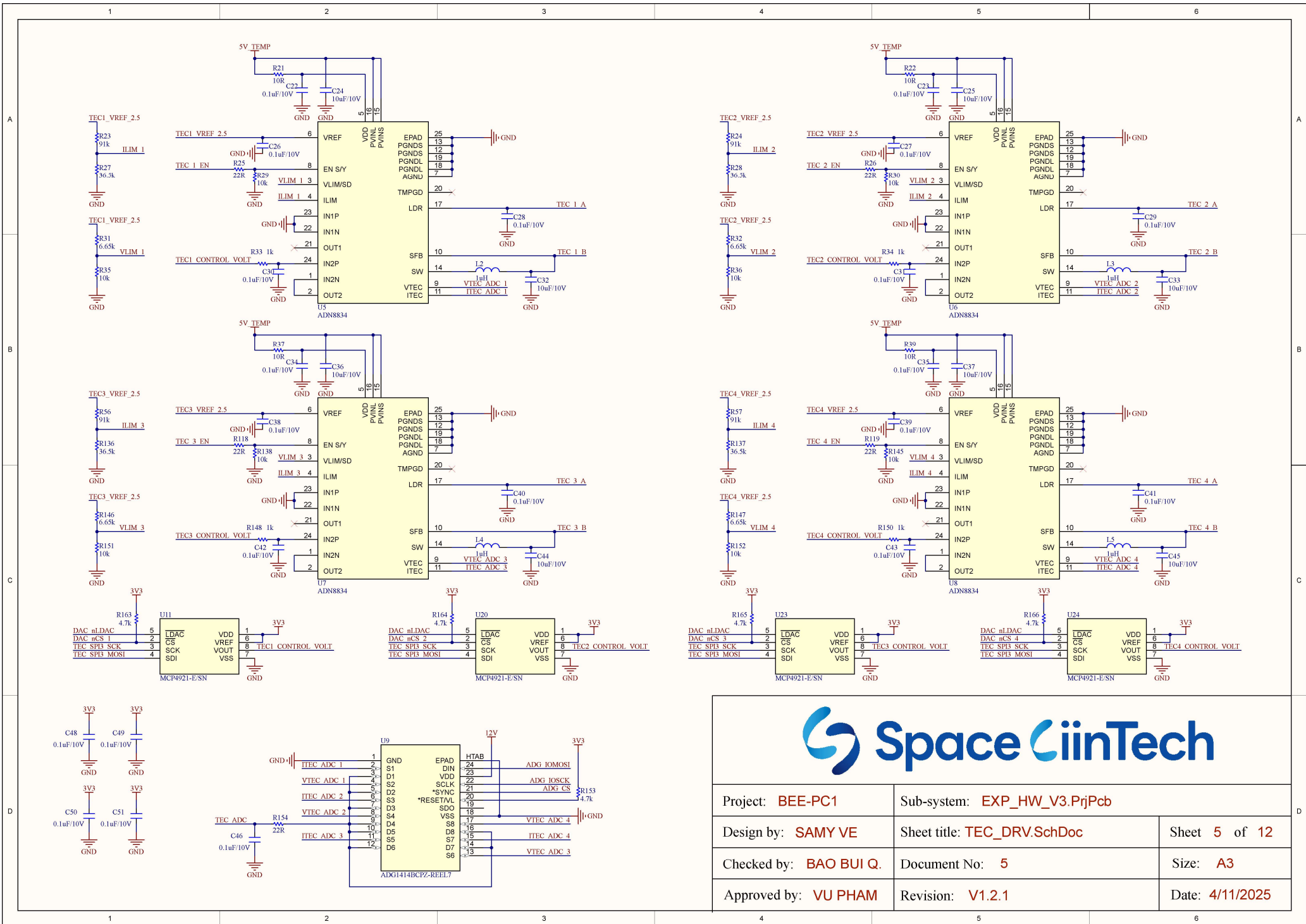


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|------------------------|-------------------------------------|-----------------|
| Project: BEE-PC1       | Sub-system: NANORACK_EXP_CONTROLLER |                 |
| Design by: SANG HUYNH  | Sheet title: HEADER_2.SchDoc        | Sheet 3 of 12   |
| Checked by: BAO BUI Q. | Document No: 3                      | Size: A3        |
| Approved by: VU PHAM   | Revision: V1.2.1                    | Date: 4/11/2025 |

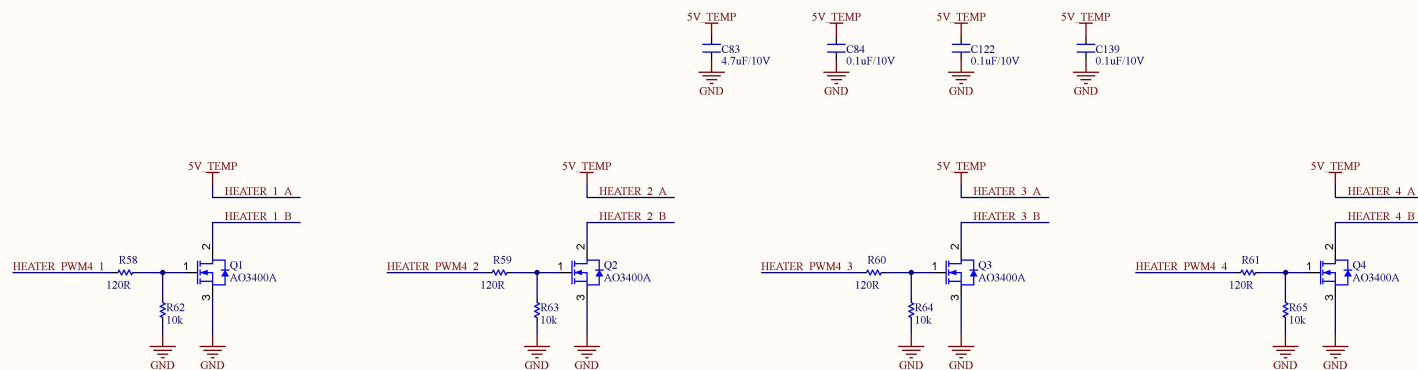


|                        |                                     |                 |
|------------------------|-------------------------------------|-----------------|
| Project: BEE-PC1       | Sub-system: NANORACK_EXP_CONTROLLER |                 |
| Design by: SANG HUYNH  | Sheet title: POWER.SchDoc           | Sheet 4 of 12   |
| Checked by: BAO BUI Q. | Document No: 4                      | Size: A3        |
| Approved by: VU PHAM   | Revision: V1.2.1                    | Date: 4/11/2025 |

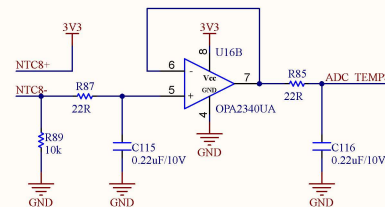
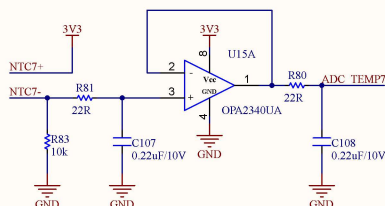
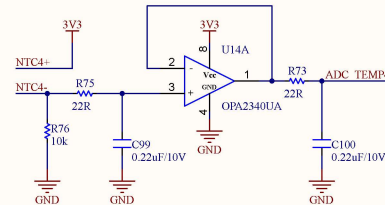
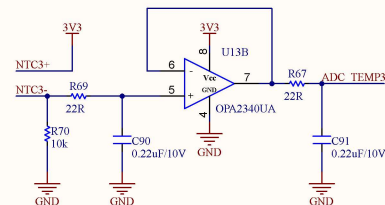
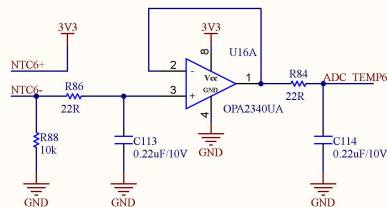
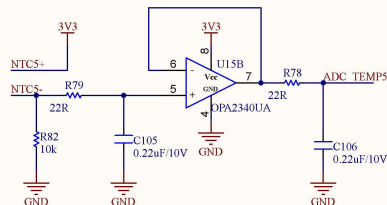
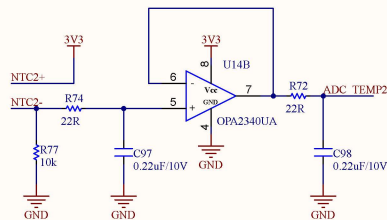
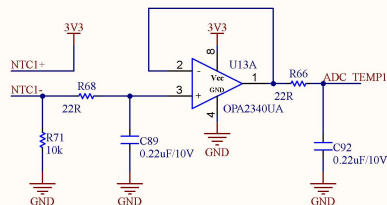
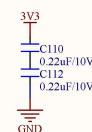
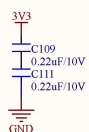
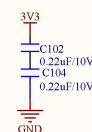
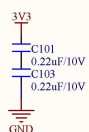
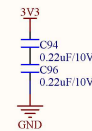
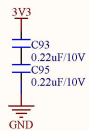
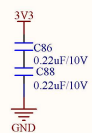
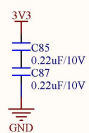




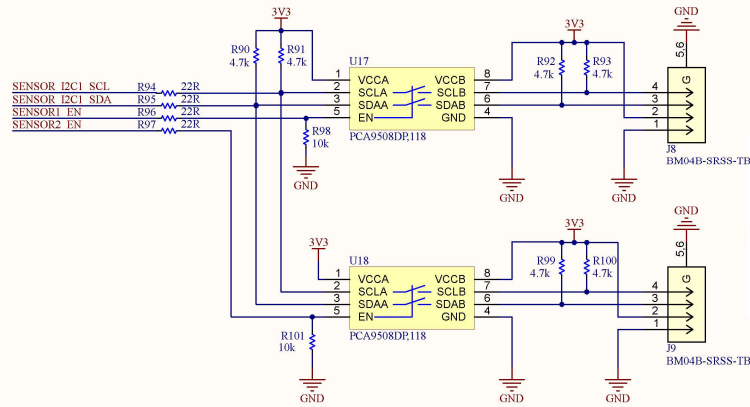
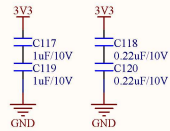
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|-------------------------------|-------------------------------------|-----------------------------|
| Project: <b>BEE-PC1</b>       | Sub-system: <b>EXP_HW_V3.PrjPcb</b> |                             |
| Design by: <b>SAMY VE</b>     | Sheet title: <b>TEC_DRV.SchDoc</b>  | Sheet <b>5</b> of <b>12</b> |
| Checked by: <b>BAO BUI Q.</b> | Document No: <b>5</b>               | Size: <b>A3</b>             |
| Approved by: <b>VU PHAM</b>   | Revision: <b>V1.2.1</b>             | Date: <b>4/11/2025</b>      |



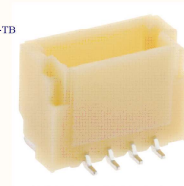
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| Project: BEE-PC1       | Sub-system: NANORACK_EXP_CONTROLLER |                 |
| Design by: SANG HUYNH  | Sheet title: HEATER.SchDoc          | Sheet 6 of 12   |
| Checked by: BAO BUI Q. | Document No: 6                      | Size: A3        |
| Approved by: VU PHAM   | Revision: V1.2.1                    | Date: 4/11/2025 |



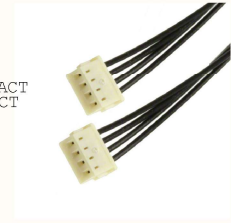
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| Project: BEE-PC1       | Sub-system: NANORACK_EXP_CONTROLLER |                 |
| Design by: SANG HUYNH  | Sheet title: NTC.SchDoc             | Sheet 7 of 12   |
| Checked by: BAO BUI Q. | Document No: 7                      | Size: A3        |
| Approved by: VU PHAM   | Revision: V1.2.1                    | Date: 4/11/2025 |



### JST Sales America Inc. BM04B-SRSS-TB



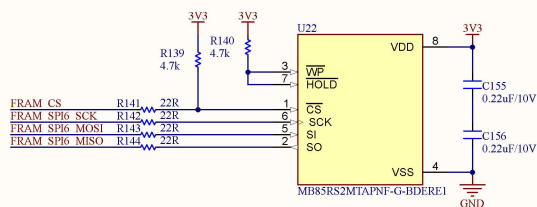
- **ELECTRICAL:**  
CURRENT RATING: 1.0A AC/DC/CONTACT  
VOLTAGE RATING: 50V AC/DC/CONTACT
- **ENVIRONMENTAL:**  
TEMPERATURE RANGE: -25°C ~ 85°C
- **PHYSICAL:**  
PITCH-MATING: 0.039" (1.00mm)
- **WIRE:**  
WIRE SIZE: 26 AWG  
MAX CURRENT: 0.1A per Wire  
ISULATION MATERIAL: PTFE
- **MATING:**  
JST Sales America Inc.  
A04SR04SR30K152B



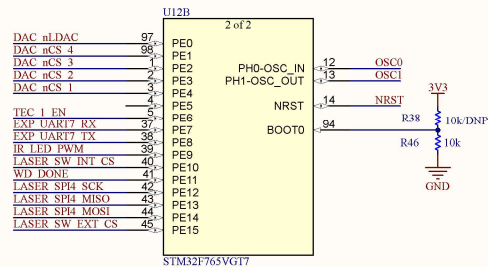
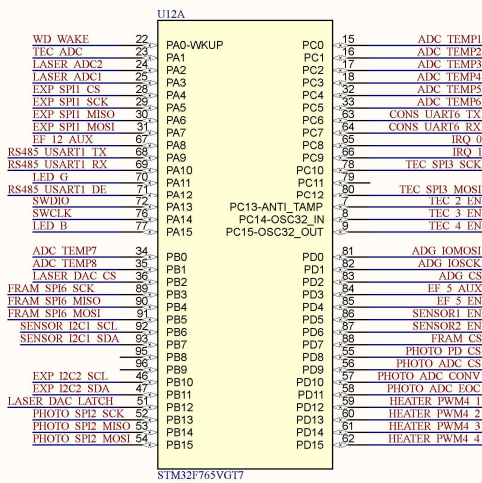
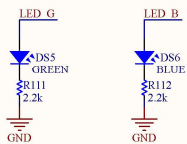
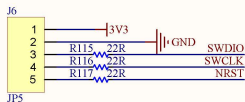
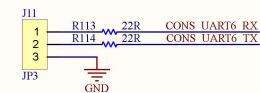
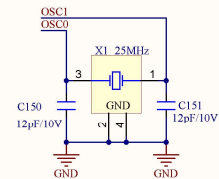
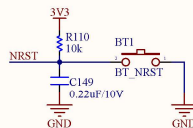
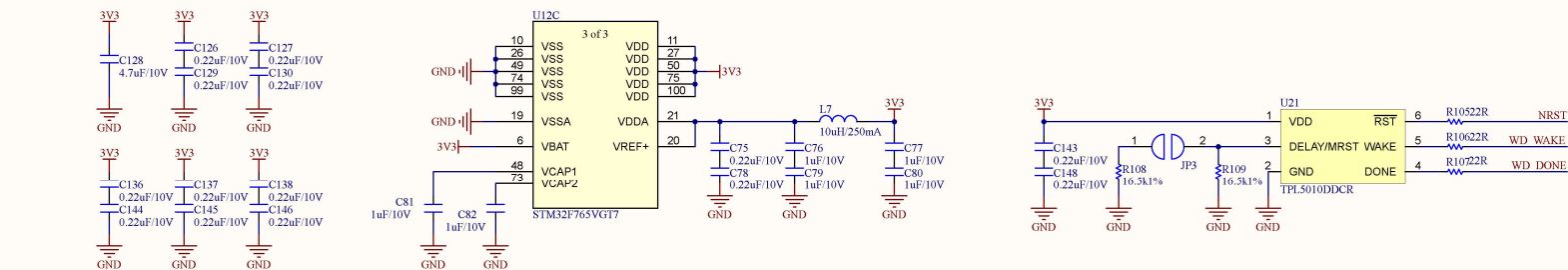
|                               |                                            |                             |
|-------------------------------|--------------------------------------------|-----------------------------|
| Project: <b>BEE-PC1</b>       | Sub-system: <b>NANORACK_EXP_CONTROLLER</b> |                             |
| Design by: <b>SANG HUYNH</b>  | Sheet title: <b>I2C_Sensor.SchDoc</b>      | Sheet <b>8</b> of <b>12</b> |
| Checked by: <b>BAO BUI Q.</b> | Document No: <b>8</b>                      | Size: <b>A3</b>             |
| Approved by: <b>VU PHAM</b>   | Revision: <b>V1.2.1</b>                    | Date: <b>4/11/2025</b>      |



Space CiiinTech

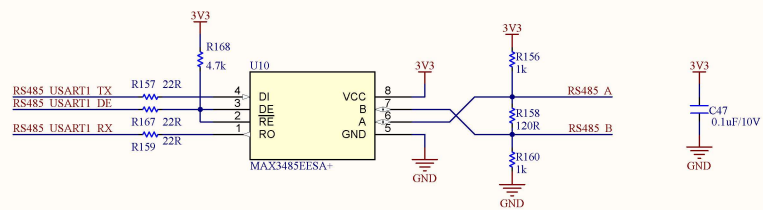


|                        |                                     |                 |
|------------------------|-------------------------------------|-----------------|
| Project: BEE-PC1       | Sub-system: NANORACK_EXP_CONTROLLER |                 |
| Design by: SANG HUYNH  | Sheet title: FRAM.SchDoc            | Sheet 10 of 12  |
| Checked by: BAO BUI Q. | Document No: 10                     | Size: A3        |
| Approved by: VU PHAM   | Revision: V1.2.1                    | Date: 4/11/2025 |



|                        |                                     |                 |
|------------------------|-------------------------------------|-----------------|
| Project: BEE-PC1       | Sub-system: NANORACK_EXP_CONTROLLER |                 |
| Design by: SANG HUYNH  | Sheet title: MCU.SchDoc             | Sheet 11 of 12  |
| Checked by: BAO BUI Q. | Document No: 11                     | Size: A3        |
| Approved by: VU PHAM   | Revision: V1.2.1                    | Date: 4/11/2025 |





|                               |                                     |                              |
|-------------------------------|-------------------------------------|------------------------------|
| Project: <b>BEE-PC1</b>       | Sub-system: <b>EXP_HW_V3.PrjPcb</b> |                              |
| Design by: <b>SAMY VE</b>     | Sheet title: <b>RS485.SchDoc</b>    | Sheet <b>12</b> of <b>12</b> |
| Checked by: <b>BAO BUI Q.</b> | Document No: <b>12</b>              | Size: <b>A3</b>              |
| Approved by: <b>VU PHAM</b>   | Revision: <b>V1.2.1</b>             | Date: <b>4/11/2025</b>       |